One learning system, three layers

PREPARE / ASTAR FORGE

Source PDF > extraction and page images > lesson review > export



DELIVER / OFFLINE SOFTWARE

Local course package > chapters and lessons > practice > local progress



POWER / PROPOSED HARDWARE

Solar panel > charge controller > rechargeable battery > device electronics

Target device blocks: controller, local storage, e-paper display and physical controls. Firmware and hardware package format remain to be specified.

Validation required: daily energy use, charging performance, battery safety, display refresh, memory use, durability, repairability and learner usability.